

Normalization Process

As indicated above, there are some difficulties with the I-Star and optimized cost index derivations. Mainly, these metrics will overstate the cost to trade because they either indicate the ultra-aggressive trading cost or incorporate an extra term (timing risk) into the estimate. For example, if the expected cost of a trade is 20 ± 50 bp the expected cost is 20 bp. If we quote a cost of 70 bp we would be overstating the true expected cost. In these cases, these derivations are often normalized based on a base time period with a value of 100. Costs are then scaled up or down based on the actual cost metric.

For example, if we set the index in Jan. 2012 = 100, then a cost index of 105 would indicate that costs are 5% higher than in Jan. 2012 and a cost index of 95 would indicate that costs are 5% lower than Jan. 2012. As we show below, the normalized cost index is extremely useful for funds to construct and customize their own historical cost index series.

The cost index normalization process is as follows:

- Start with the trading cost calculation such as that in [Equation 12.1](#) or [Equation 12.2](#) and compute this cost metric over the required period.
- Select the base time period. For illustrative purposes we set Jan. 2012 to be our base month (e.g., Jan. 2012 = 100).
- Compute the normalized cost in each period following:

$$\text{Normalized Cost}(t) = \frac{I^*(t)}{I^*(\text{Jan } 2012)} \quad (12.3)$$

[Figure 12.2](#) illustrates the normalization process using the I-Star cost calculation for large cap stocks. We compute the I-Star cost in each month based using an order size of 10% ADV and then normalize based on Jan. 2012 as the base year. That is, Jan. 2012 = 100. The normalized cost index can be constructed for different sizes or by dollar value or share quantity.

Portfolio managers are often interested in the cost of trading different size orders such as 5% or 25% ADV or trading via a different strategy than was used in the historical index calculation. These costs could vary greatly from the cost computed for 10% ADV. In these situations, managers have a few options available. They could recreate the cost index for the specified order size and market conditions. But computing the cost index for the desired order size over the required time period could become a very time consuming exercise, especially if the portfolio manager wishes to evaluate several different order sizes and various different strategies, as well as a large number of potential stocks.